

Taking Flight

Flying is incredibly powerful in combat and should be a difficult task to accomplish.

To determine whether a creature can fly with wings, use Flight Power Ratio (FPR). Wing Area is calculated from wing sizes. Total Lifted Weight includes the character's modified body mass (which may be reduced by traits like *Hollow Bones*) plus all carried equipment. If FPR is **1.0 or higher**, sustained flight is possible. The greater the FPR, the more efficient the flight: values around 1.2 allow average flight, 1.6 - 2.0 support full maneuverability, and values over 2.0 unlock exceptional flight. Consider encumbrance, traits, gear, and skills.

For ease of recalculating weight, note the Lift Power separately from FPR. Ex. Fly SPD 30ft (LP 106.6. FPR 53.3)

Flying speed is the Size SPD × FPR. FPR can not exceed 3 for base SPD calculations. This cap only applies to SPD calculations, and may have different caps for different Abilities. SPD may still be increased through traits, skills, etc.

FPR = Lift Power / Total Lifted Weight

Lift Power = Wing Area × Flap Power

Wingspan = Measured from tip to tip when fully extended.

Wing Area = (Wingspan²) / 3. Round to the nearest hundredth (0.01) if the result is under 10, tenth if under 100, and the whole number if over 100.

Flap Power = (BDY + SPT) × Wing Multiplier

Wing Types

- Bird-like: × 5
- Bat-like: × 4
- Insect-like: × 3
- Glider: Use Flap = 1 for calculation only. Gliders do not use flap-based actions.

Traits may increase Wing Multiplier, reduce Lifted Weight, or bypass FPR restrictions.

Note: Gliders rely on air currents or altitude to initiate movement.

Total Lifted Weight = Character Weight + Weight of carried items or characters. Affected by traits and skills.

Interpreting FPR

FPR	Result
< 1.0	Cannot sustain flight; may glide or attempt Takeoff at +1 AP and double SP cost.
1.0–1.2	Can Takeoff and Land normally; Maneuver, Hover and Soar at +2 SP cost. Soaring loses 20ft instead of 5ft.
1.2–1.5	Can Takeoff, Land, Maneuver, Hover, and Soar normally. Sustained flight is viable.
1.6–2.0	May use all Flight Move Actions without penalty.
> 2.0	Take-off -1 AP -1 SP. Never take fall damage from landing unless the flight mechanism is disabled.

Flight Move Actions

Action	Notes
Takeoff	Base 3 AP 3 SP; -1AP with running start.
Landing	Free if only 1 move action taken. SPT or Flying skill vs DC 5 × moves taken. Failure results in ½ fall damage. Critical failure results in full fall damage. Calculate fall damage based on full SPD moved this round.
Maneuver	Turning 90 degrees or more per move action costs an additional 1 AP and SP.
Soar	When altitude Tier 1 or greater movement with less than a 90 degree turn costs -1 AP and -1 SP. Lose 5 ft of altitude per move while soaring.
Ascend	½ SPD when moving up at a 45 degree or higher angle from the horizon.
Descend	SPD × 2, Costs 0 AP and SP. Add 5 ft of base speed per turn spent Descending.
Hover	SP cost is 4 / FPR (rounded down) if no move is taken while flying. If SP cost would be below 1, or if character doesn't have enough SP, fall at a rate of 20ft per turn. If partial SP is spent, fall at half the speed.
Glide	Moving is fixed to SPD / 2 but costs 0 AP and SP. Must move full speed per turn. Can not take additional move actions while gliding if FPR <1.

See Altitude Modifiers table for tier-based penalties to Flight Checks and SP costs.
See Falling rules under General Damage.

Wing Area Reference Table

Animal	Body Mass (lbs)	Wingspan (ft)	Estimated Wing Area (ft²)
Dragonfly	0.02	0.5	0.08
Sparrow	0.04	0.8	0.21
Giant Hornet	0.1	1.2	0.48
Pigeon	0.70	2.0	1.33
Crow	1.00	2.8	2.61
Hawk	2.00	4.0	5.33
Bald Eagle	10.00	7.5	18.75
Vulture	15.00	9.0	27.00
Albatross	20.00	11.0	40.33
Human-Sized Flyer	180.00	18.0	108.00
Wyvern	600.00	32.0	341.33
Dragon	1800.00	50.0	833.33

Examples

Dragonfly (Tiny)

Size SPD: 10
Wingspan: 0.5 ft
Wing Area: $(0.5 \times 0.5) / 3 = 0.083 \text{ ft}^2$
BDY + SPT: $1 + 2 = 3$
Wing Type: Insect (x 3)
Flap Power: $3 \times 3 = 9$
Total Lifted Weight: 0.02 lbs
FPR: $(0.083 \times 9) / 0.02 = 37.5$
SPD Calculation: 10×3 (capped) = 30 ft

Hawk (Tiny)

Size SPD: 10
Wingspan: 4
Wing Area: $(4 \times 4) / 3 = 5.33 \text{ ft}^2$
BDY + SPT: 4
Wing Type: Bird (x 5)
Flap Power = $4 \times 5 = 20$
Total Lifted Weight: 2 lbs
FPR = $(5.33 \times 20) / 2 = 53.3$
SPD = 10×3 (capped) = 30 ft

Harpy (Medium)

Size SPD: 20
Wingspan: 18 ft
Wing Area: $(18 \times 18) / 3 = 108 \text{ ft}^2$
BDY + SPT: $4 + 4 = 8$
Wing Type: Bird-like x 5
Flap Power = $8 \times 5 = 40$
Total Lifted Weight = $180 + 30 = 210 \text{ lbs}$
(180 Body Mass, 30 Carried)
FPR = $(108 \times 40) / 210 = 20.5$
SPD = 20×3 (capped) = 60 ft/round

Dragon (Huge)

Size SPD: 30
Wingspan: 50
Wing Area: $(50 \times 50) / 3 = 833 \text{ ft}^2$
BDY + SPT: 10
Wing Type: Bat (x 4)
Flap Power = $10 \times 4 = 40$
Total Lifted Weight: 1800 lbs
FPR = $(833 \times 40) \div 1800 = 18.5$
SPD = 30×3 (capped) = 90 ft

Flying Altitude Modifiers

Tier	Altitude	Air Density	Penalty Example
Tier 0	0–4,999 ft	100%	No penalty
Tier 1	5,000–9,999 ft	~83%	–1 to Flight checks, +1 SP to Sustain
Tier 2	10,000–14,999 ft	~70%	–2 Flight Checks, –5ft Speed, +1 additional SP to sustain
Tier 3	15,000–19,999 ft	~60%	–3 Flight Checks, –10ft Speed, +1 additional SP to Sustain
Tier 4	20,000–24,999 ft	~53%	Must roll CON/SPT to breathe
Tier 5+	25,000+ ft	≤50%	Flight impossible without magic/trait